Naksh Sharma

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Education

International Institute of Information Technology, Bangalore **Expected Graduation: 2029**
Bachelor of Technology in Artificial Intelligence and Data Science *CGPA 3.13/4.0 (1st Year)*

Experience

Aivancity School for Technology, Business & Society **Paris, France**
AI & Big Data Study Abroad Program — FairSplit Team (4-person team) *Jun 2026*

- Studied **Big Data, Artificial Intelligence, and business**, alongside the ethical and legal dimensions of AI, including EU rules and the legitimate use of AI systems.
- Our four-person team developed **FairSplit**, a property-valuation project for divorce settlements designed to help parties and their advisers understand a home's estimated value through a clear, evidence-based process.
- As the primary ML contributor, built and compared **ElasticNet and CatBoost** models using the Kaggle Ames Housing benchmark; the selected model achieved a five-fold cross-validated **RMSLE of 0.113**.
- Developed the Python valuation workflow to process property attributes, generate estimates, explain key value drivers with **SHAP**, and produce structured PDF reports; presented the project as the program capstone to complete the residency and receive our certificates.

Projects

InsightGraph — Local Knowledge-Graph RAG System | *Python, NetworkX, ChromaDB, Ollama* **Jul 2026**

- Built an ingestion pipeline that chunks documents, extracts entities and relations via **LLM prompting**, deduplicates them with fuzzy matching, and writes to a **NetworkX** graph store and a **ChromaDB** vector store.
- Designed a **hybrid retrieval** layer fusing vector search with graph-neighbor lookups to answer multi-hop, multi-document relationship questions with source citations, running fully on-device via **Llama 3.1 8B** on Ollama.
- Added a Streamlit chat interface and validated the system with **39 pytest tests** plus a gold-standard Q&A evaluation suite.

RaceCraft — Counterfactual F1 Race-Strategy Simulator | *Python, FastF1, Streamlit* **Oct 2026**

- Built a **Monte Carlo** simulator comparing pit-now, extend, and compound-change strategies as a distribution of finish positions, modelling tyre degradation, pit loss, traffic, and safety cars across **all 24 2026 circuits**.
- Built an **all-car digital twin** with opponent pit policies and sampled VSC/safety-car events, so each counterfactual plays out against realistic competitor behaviour.
- Added OpenF1/FastF1 data ingestion with **leak-safe feature pipelines**, an OCR reader for timing screenshots, a Streamlit dashboard, **13 pytest tests**, and CI.

SML NLP — News Classification | *Python, PyTorch, Transformers, scikit-learn* **Apr 2026**

- Built a staged text-classification pipeline on an **8,000-example AG News sample**, comparing unigram/bigram TF-IDF features with contextual embeddings from pretrained transformer encoders.
- Evaluated classical classifiers on a held-out test set; a **DistilBERT mean-pooled embedding + RBF-SVM** achieved **89.7% macro-F1**, versus **88.3%** for the best bigram TF-IDF baseline.
- Added reproducible experiment configuration, encoder comparisons, per-class evaluation, and generated result tables and visualizations.

Technical Skills

Languages: Python, SQL, C++, C
AI / ML: Machine Learning, Neural Networks, scikit-learn, CatBoost, SHAP, RAG, Isolation Forest
Technologies/Frameworks: Pandas, NumPy, FastAPI, React, PostgreSQL, Docker, NetworkX, ChromaDB, Streamlit
Tools: Git/GitHub, Google Colab, Visual Studio Code

Relevant Coursework

- Data Structures & Algorithms
- Database Management
- Operating Systems
- Computer Architecture
- Machine Learning
- Neural Networks

Achievements

- Team **semifinalist** at **Novo Nordisk's national hackathon** with Agentic Prompt Guard.
- Secured **AIR 8045** with **99.48 percentile** in JEE Mains.
- Scored **96%** in Class 12 (Sri Chaitanya Junior College).
- Awarded a **Certificate of Completion** by **Aivancity School for Technology, Business & Society**, Paris, for completing a 2-week intensive residency in AI, Big Data, Business, and Ethics.